

Solana Esports Platform

Monetized Matchmaking, MPC Wallets & Prize Distribution

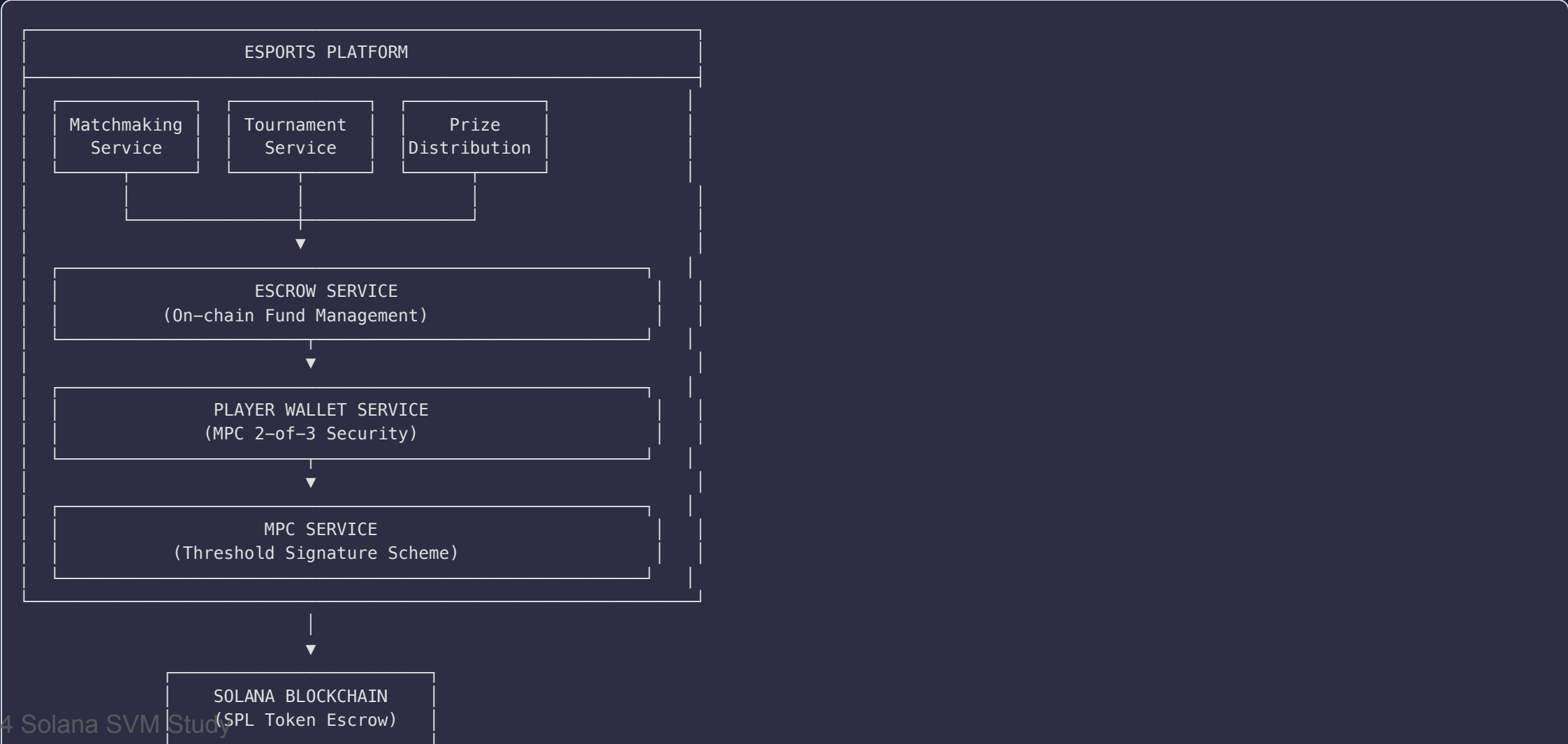
Production-Grade Implementation

Agenda

1. **System Overview** - Architecture & Components
2. **Service Layer Architecture** - Core Services Deep Dive
3. **MPC Wallet Integration** - Secure Player Wallets
4. **Matchmaking System** - Entry Fees & Escrow
5. **Prize Distribution** - Calculations & Strategies
6. **Tournament Management** - Bracket Generation
7. **Security Practices** - Production Considerations
8. **Implementation Details** - Code Walkthrough

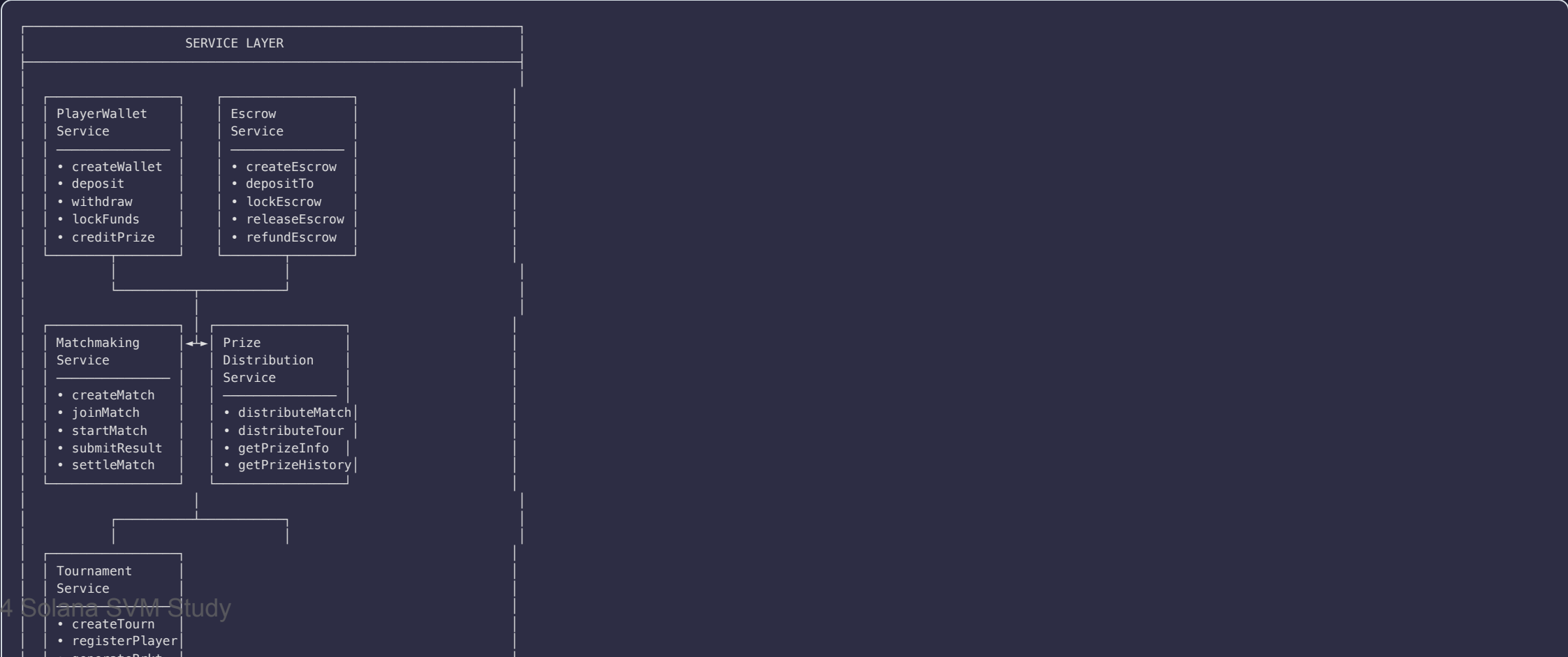


System Architecture Overview



Service Layer Architecture

Five Core Services



PlayerWalletService

MPC-Secured Wallet Management

```
@Injectable()
export class PlayerWalletService {
  // Creates MPC wallet with 2-of-3 threshold scheme
  async createWallet(request: CreatePlayerWalletRequest) {
    const mpcWallet = await this.mpcService.createMpcWallet({
      name: `Player Wallet - ${playerId}`,
      thresholdScheme: ThresholdScheme.TSS_2_3,
      participants: [
        { participantId: `player_${playerId}`, ... }, // Device key
        { participantId: 'platform_signer', ... }, // Platform HSM
        { participantId: 'recovery_service', ... }, // Cold storage
      ],
    });
    // Store wallet with default limits
    return this.walletRepository.create({
      playerId,
      mpcWalletId: mpcWallet.walletId,
      dailyWithdrawalLimit: DEFAULT_DAILY_LIMIT, // 10 SOL
      status: PlayerWalletStatus.ACTIVE,
    });
  }
}
```



PlayerWalletService Methods

Method	Purpose	Key Logic
createWallet	Initialize MPC wallet	2-of-3 threshold scheme
deposit	Record external deposit	Verify on-chain signature
withdraw	MPC-signed withdrawal	Rate limit + daily cap
lockFunds	Reserve for escrow	Move to lockedBalance
unlockFunds	Release from escrow	Restore to availableBalance
creditPrize	Award winnings	Direct credit + audit log

Balance Tracking

```
interface WalletBalance {
  availableBalance: string; // Spendable
  lockedBalance: string;    // In active matches
  totalDeposited: string;   // Lifetime in
  totalWithdrawn: string;   // Lifetime out
}
```

EscrowService

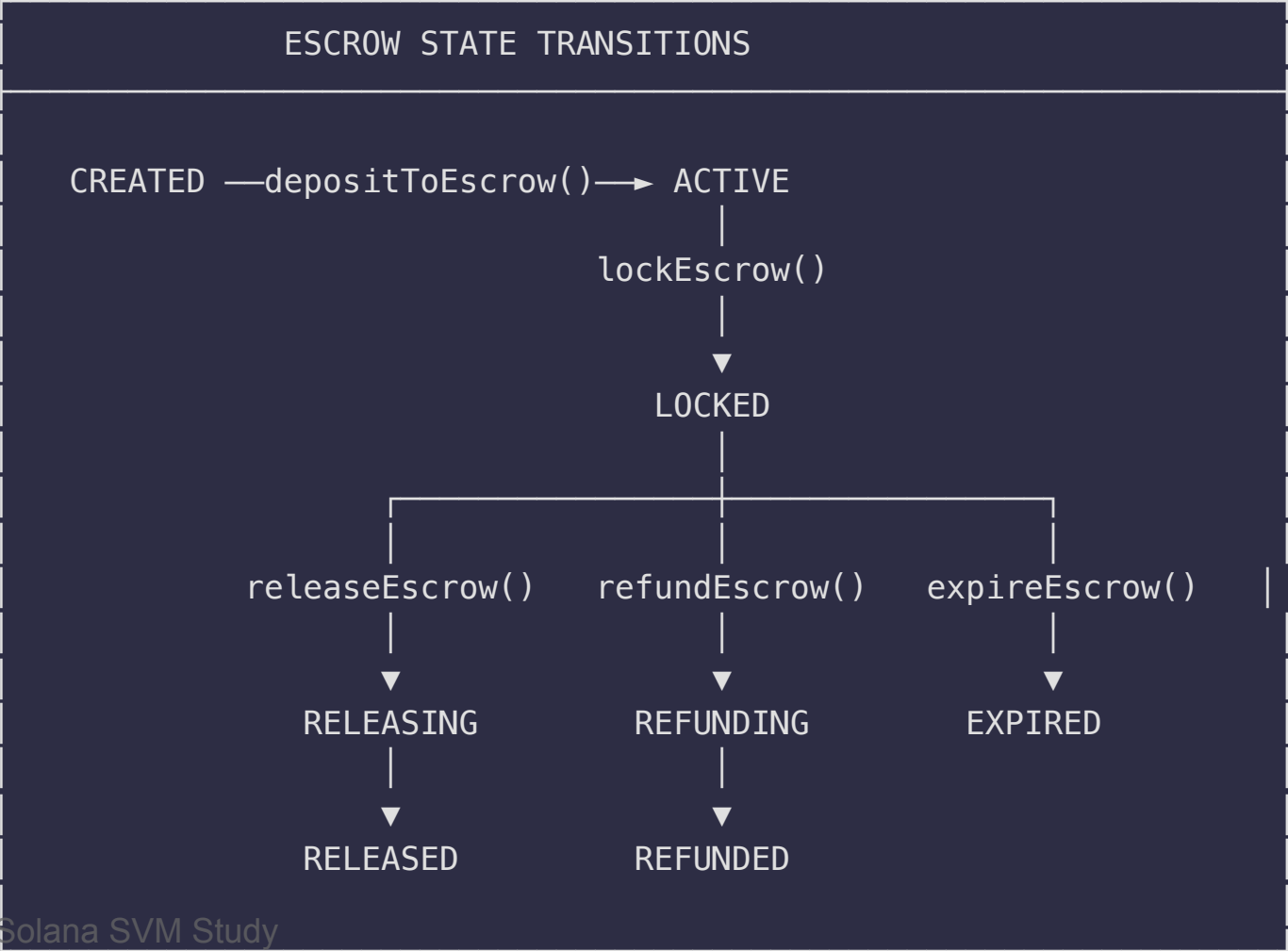
Trustless Fund Management

```
@Injectable()
export class EscrowService {
  // Generate deterministic escrow address (PDA simulation)
  private generateEscrowAddress(sourceType, sourceId): string {
    const seed = `${sourceType}:${sourceId}`;
    return createHash('sha256')
      .update(seed)
      .digest('hex')
      .slice(0, 44);
  }

  async createEscrow(request: CreateEscrowRequest) {
    const escrowAddress = this.generateEscrowAddress(
      request.sourceType,
      request.sourceId
    );
    return this.escrowRepository.create({
      escrowId: `escrow_${randomBytes(8).toString('hex')}`,
      escrowAddress,
      sourceType,      // MATCH or TOURNAMENT
      sourceId,        // match_xxx or tournament_xxx
      platformFeePercent: 5.0,
      status: EscrowStatus.CREATED
    });
  }
}
```



EscrowService State Machine



MatchmakingService

Core Match Operations

```
@Injectable()
export class MatchmakingService {
  async createMatch(request: CreateMatchRequest): Promise<Match> {
    // 1. Validate entry fee bounds
    const minFee = BigInt(process.env.MIN_ENTRY_FEE || '1000000');
    const maxFee = BigInt(process.env.MAX_ENTRY_FEE || '1000000000000');

    // 2. Create escrow account
    const escrow = await this.escrowService.createEscrow({
      sourceType: EscrowSourceType.MATCH,
      sourceId: matchId,
      platformFeePercent,
    });

    // 3. Create match entity
    return this.matchRepository.create({
      matchId,
      gameType,
      entryFee,
      status: MatchStatus.CREATED,
      escrowAddress: escrow.escrowAddress,
```



MatchmakingService - Join Flow

```
async joinMatch(request: JoinMatchRequest): Promise<MatchParticipant> {  
  const match = await this.getMatchById(matchId);  
  
  // 1. Validate match is joinable  
  if (!match.isJoinable()) {  
    throw new BadRequestException('Match not accepting players');  
  }  
  
  // 2. Get player wallet and verify balance  
  const wallet = await this.playerWalletService.getWallet(playerId);  
  if (BigInt(wallet.availableBalance) < BigInt(match.entryFee)) {  
    throw new BadRequestException('Insufficient balance');  
  }  
  
  // 3. Lock entry fee in player wallet  
  await this.playerWalletService.lockFunds({  
    playerId,  
    amount: match.entryFee,  
    reference: matchId,  
  });  
  
  // 4. Deposit to escrow  
  await this.escrowService.depositToEscrow({  
    escrowId,  
    walletId: wallet.id,  
    amount: match.entryFee,  
  });  
  
  // 5. Update prize pool  
  match.prizePool = (BigInt(match.prizePool) + BigInt(entryFee)).toString();  
}
```

PrizeDistributionService

Prize Calculation Strategy

```
@Injectable()
export class PrizeDistributionService {
  async distributeMatchPrizes(match, participants): Promise<PrizeDistribution> {
    // 1. Get escrow balance
    const escrowBalance = await this.escrowService.getEscrowBalance(escrowId);
    const totalPrizePool = BigInt(escrowBalance.currentBalance);

    // 2. Calculate platform fee (basis points for precision)
    const platformFee = (totalPrizePool *
      BigInt(Math.round(match.platformFeePercent * 100))) /
      BigInt(10000);

    // 3. Calculate distributable amount
    const distributableAmount = totalPrizePool - platformFee;

    // 4. Calculate individual distributions
    const distributions = this.calculateMatchDistributions(
      match, participants, distributableAmount
    );

    // 5. Release escrow and credit wallets
    await this.escrowService.releaseEscrow({ escrowId, distributions });

    for (const dist of distributions) {
      await this.playerWalletService.creditPrize(
        dist.playerId, dist.amount, match.matchId
      );
    }
  }
}
```



Prize Calculation Methods

Match Distribution Strategies

```
private calculateMatchDistributions(  
  match: Match,  
  participants: MatchParticipant[],  
  distributableAmount: bigint  
): PrizeCalculation[] {  
  const distributions: PrizeCalculation[] = [];  
  
  // Strategy 1: Duel (1v1) - Winner Takes All  
  if (match.gameType === 'duel' && match.result?.winnerIds?.length === 1) {  
    const winner = participants.find(p => p.playerId === winnerId);  
    distributions.push({  
      walletId: winner.walletId,  
      playerId: winner.playerId,  
      placement: 1,  
      amount: distributableAmount.toString(),  
      percentage: 100,  
    });  
  }  
  
  // Strategy 2: Team/Multi-Winner - Equal Split  
  else if (match.result?.winnerIds?.length > 1) {  
    const winnerCount = match.result.winnerIds.length;  
    const prizePerWinner = distributableAmount / BigInt(winnerCount);  
    // ... distribute equally  
  }  
}
```



Tournament Prize Structure

Configurable Payout Tiers

```
private calculateTournamentDistributions(
  tournament: Tournament,
  registrations: TournamentRegistration[],
  distributableAmount: bigint
): PrizeCalculation[] {
  const distributions: PrizeCalculation[] = [];

  // Sort by final placement
  const ranked = registrations
    .filter(r => r.finalPlacement != null)
    .sort((a, b) => a.finalPlacement - b.finalPlacement);

  // Apply prize structure
  for (const prizeEntry of tournament.prizeStructure) {
    const registration = ranked.find(
      r => r.finalPlacement === prizeEntry.place
    );

    if (registration) {
      // Support both percentage and fixed amounts
      const amount = prizeEntry.fixedAmount
        ? BigInt(prizeEntry.fixedAmount)
        : (distributableAmount * BigInt(prizeEntry.percentage)) / 100n;

      distributions.push({
        playerId: registration.playerId,
        placement: prizeEntry.place,
        amount: amount.toString(),
        percentage: prizeEntry.percentage,
      });
    }
  }
}
```



Fee Calculation Deep Dive

Platform Fee with Basis Points

```
// Why basis points? Precision without floating point errors

// BAD: Floating point multiplication
const fee = totalPool * 0.05; // May have rounding errors

// GOOD: Basis points (100 basis points = 1%)
const platformFee = (totalPrizePool *
  BigInt(Math.round(platformFeePercent * 100))) /
  BigInt(10000);

// Example: 5% fee on 2 SOL (2,000,000,000 lamports)
// platformFeePercent = 5.0
// Math.round(5.0 * 100) = 500 (basis points)
// (2,000,000,000 * 500) / 10000 = 100,000,000 lamports
// = 0.1 SOL platform fee
// Winner receives: 1.9 SOL
```

TournamentService

Bracket Generation Algorithm

```
@Injectable()
export class TournamentService {
  async generateBracket(tournamentId: string): Promise<Tournament> {
    const tournament = await this.getTournamentById(tournamentId);
    const registrations = await this.registrationRepository.find({
      where: { tournamentId: tournament.id },
    });

    // Seed players (by skill rating or random)
    const seededPlayers = this.seedPlayers(registrations);

    // Generate bracket based on type
    let bracket: BracketNode[];
    switch (tournament.bracketType) {
      case BracketType.SINGLE_ELIMINATION:
        bracket = this.generateSingleElimination(seededPlayers);
        break;
      case BracketType.DOUBLE_ELIMINATION:
        bracket = this.generateDoubleElimination(seededPlayers);
        break;
      case BracketType.ROUND_ROBIN:
        bracket = this.generateRoundRobin(seededPlayers);
        break;
    }

    // Create match entities for each bracket slot
    await this.createBracketMatches(tournament, bracket);
  }
}
```

Single Elimination Bracket

Seeded Matchups (Standard 1v8, 2v7, etc.)

```
private generateSingleElimination(
  players: TournamentRegistration[]
): BracketMatch[] {
  const n = players.length;
  const rounds = Math.ceil(Math.log2(n));
  const bracket: BracketMatch[] = [];

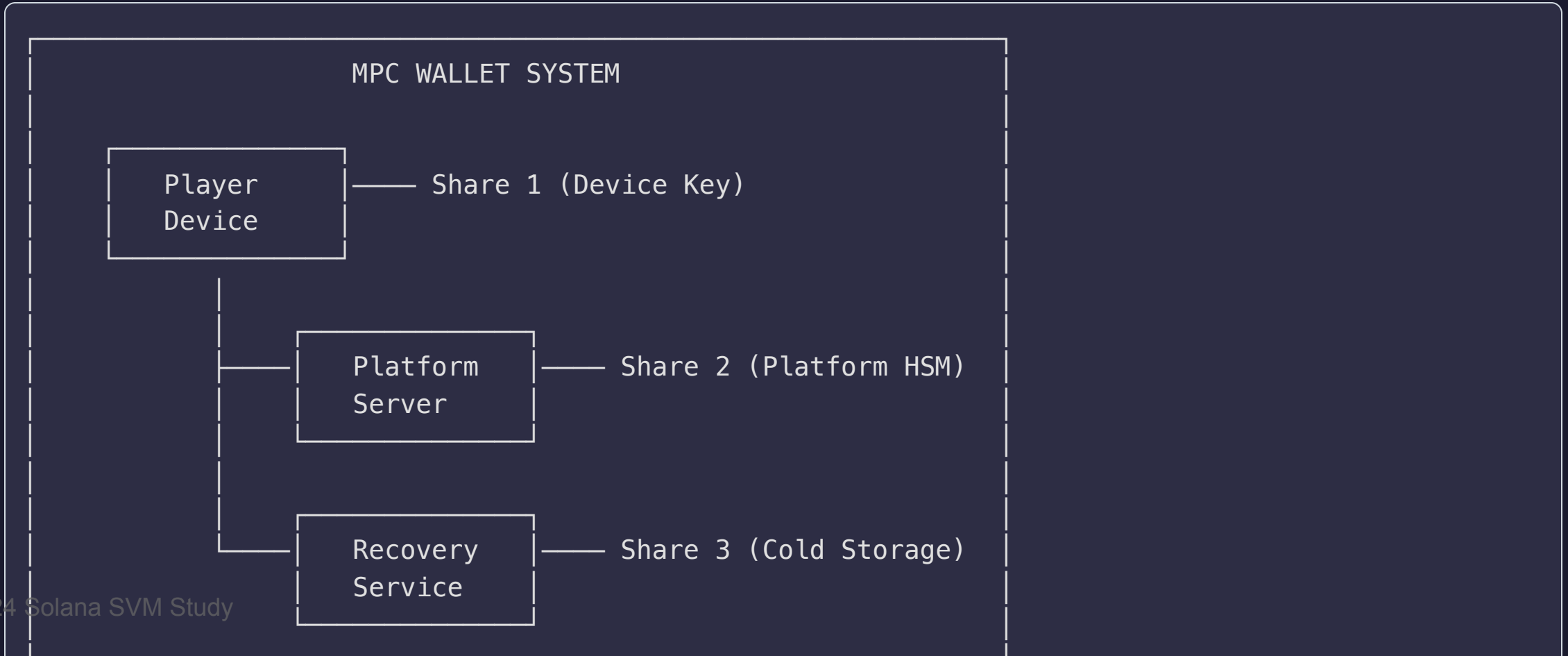
  // First round matchups (seeded)
  // Seed 1 vs Seed 8, Seed 2 vs Seed 7, etc.
  for (let i = 0; i < n / 2; i++) {
    const highSeed = players[i];
    const lowSeed = players[n - 1 - i];

    bracket.push({
      round: 1,
      position: i,
      player1Id: highSeed?.playerId,
      player2Id: lowSeed?.playerId,
      nextMatchPosition: Math.floor(i / 2),
    });
  }

  // Generate placeholder matches for subsequent rounds
  for (let round = 2; round <= rounds; round++) {
    const matchesInRound = Math.pow(2, rounds - round);
    for (let i = 0; i < matchesInRound; i++) {
      bracket.push({ round, position: i, player1Id: null, player2Id: null });
    }
  }
}
```


MPC Wallet Architecture

Multi-Party Computation (2-of-3 Threshold)





MPC Benefits for Gaming

Feature	Traditional Wallet	MPC Wallet
Single Point of Failure	✗ Yes	✓ No
Key Recovery	✗ Complex	✓ Built-in
Account Takeover Risk	✗ High	✓ Mitigated
User Experience	✗ Seed Phrases	✓ Seamless
Platform Control	✗ None	✓ Fraud Prevention

Key Advantages:

- **No seed phrase exposure** to players
- **Platform co-signing** prevents fraud
- **Recovery service** for lost devices
- **Rate limiting** on withdrawals



Player Wallet Entity

```
@Entity('player_wallets')
export class PlayerWallet {
  @PrimaryGeneratedColumn('uuid')
  id: string;

  @Column({ unique: true })
  playerId: string;

  @Column()
  mpcWalletId: string; // Link to MPC system

  @Column()
  publicKey: string; // Solana address

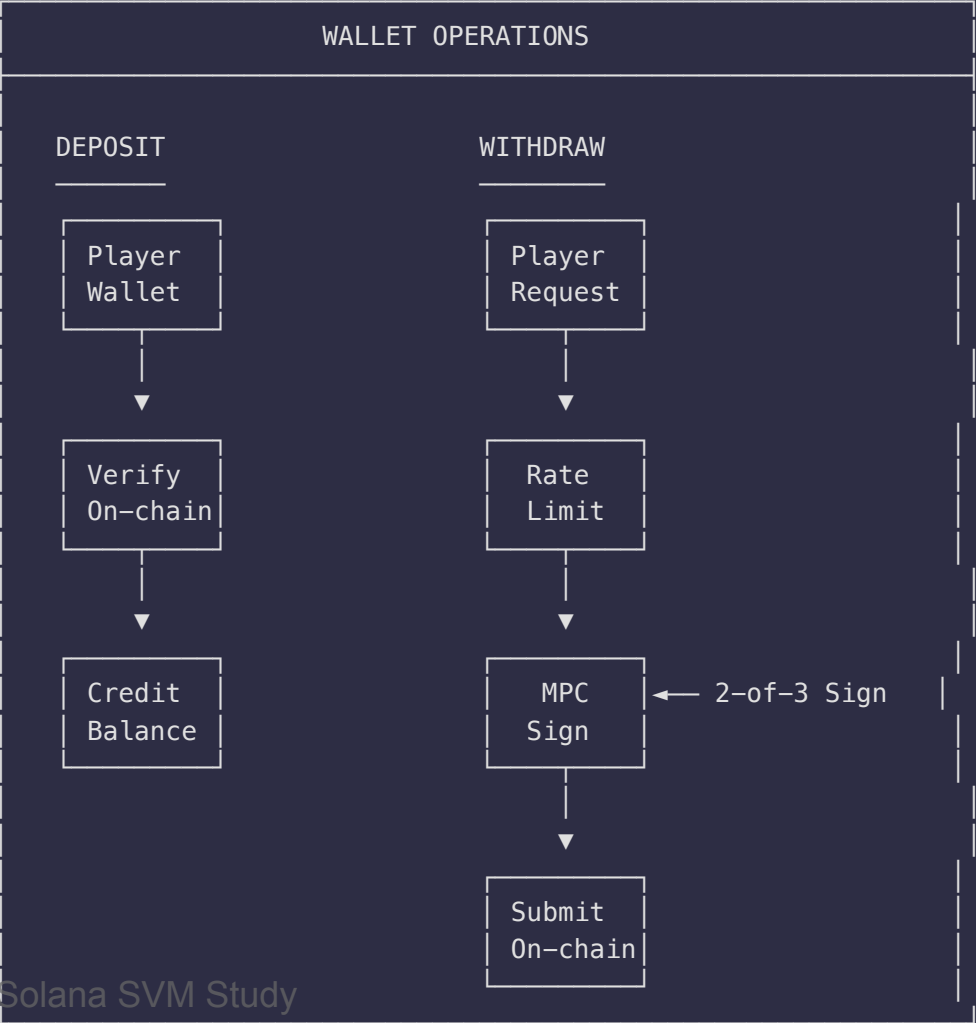
  @Column({ type: 'decimal', precision: 20, scale: 0, default: '0' })
  availableBalance: string; // Lamports

  @Column({ type: 'decimal', precision: 20, scale: 0, default: '0' })
  lockedBalance: string; // In escrow

  @Column({ type: 'decimal', precision: 20, scale: 0, default: '0' })
  dailyWithdrawalLimit: string;

  @Column({ type: 'enum', enum: PlayerWalletStatus })
  status: PlayerWalletStatus;
}
```

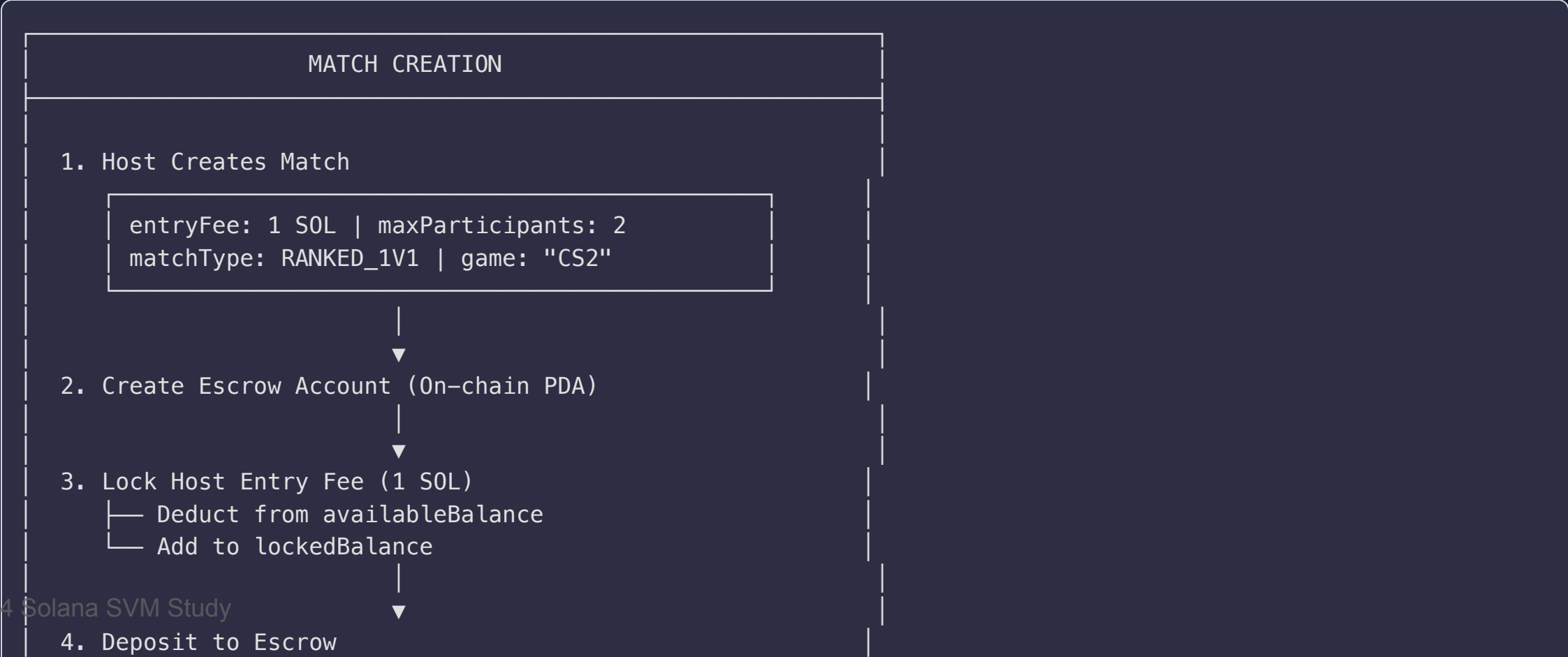
Wallet Operations Flow





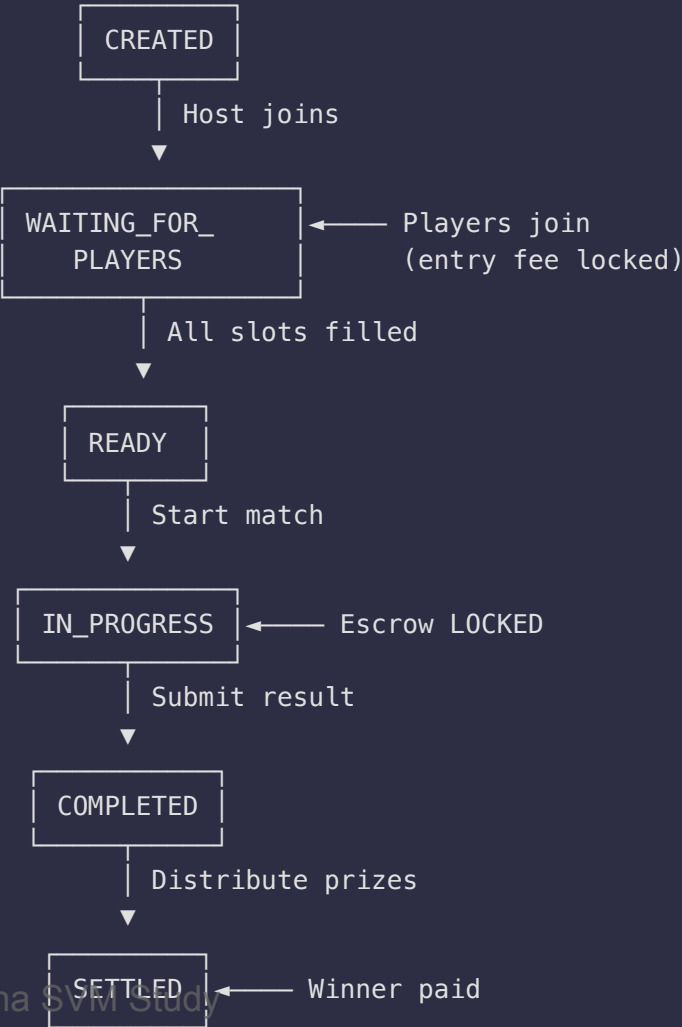
Matchmaking System

Entry Fee & Escrow Flow





Match Lifecycle



Prize Distribution Logic

Winner Takes Pool (minus platform fee)

```
async distributeMatchPrizes(matchId: string) {
  const match = await this.matchRepository.findOne({
    where: { id: matchId },
    relations: ['escrow', 'participants'],
  });

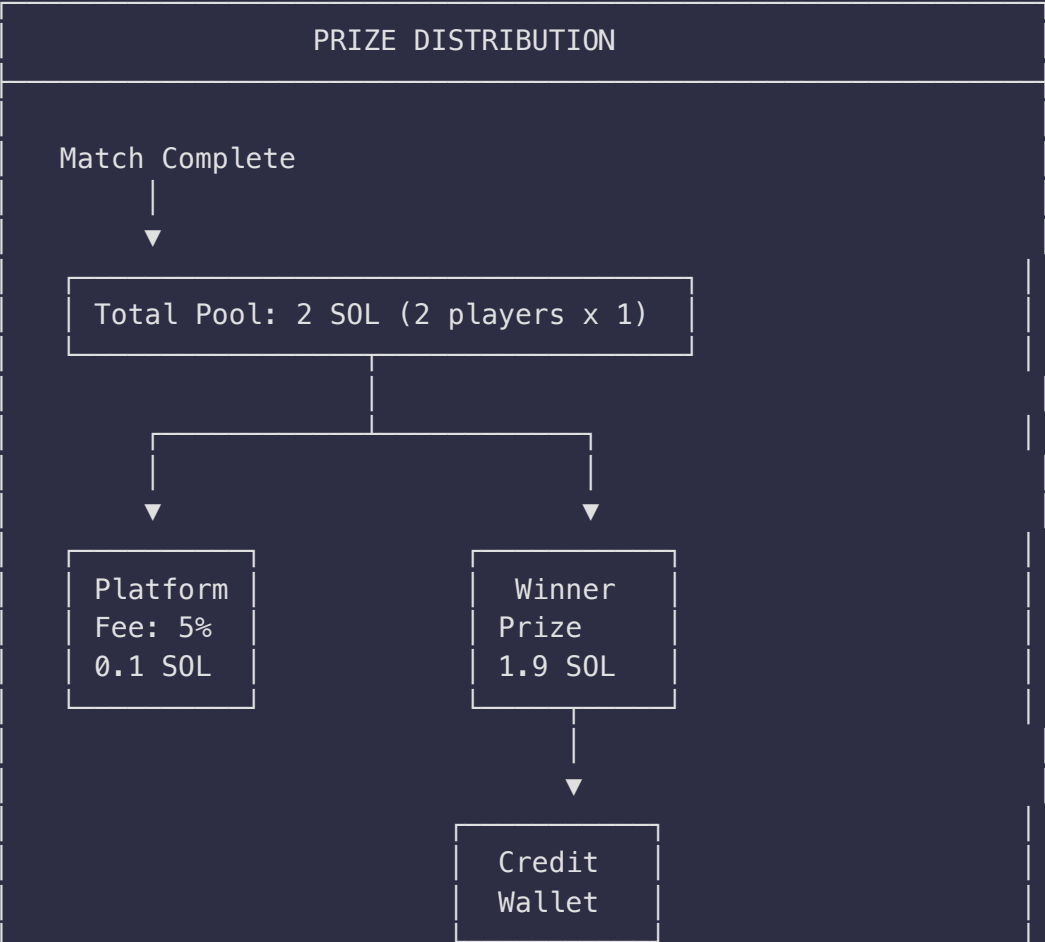
  // Calculate prize pool
  const totalPool = BigInt(match.entryFee) *
    BigInt(match.participants.length);

  // Deduct platform fee (e.g., 5%)
  const platformFee = totalPool * BigInt(match.platformFeePercentage) / 100n;
  const winnerPrize = totalPool - platformFee;

  // Credit winner's wallet
  await this.walletService.creditPrize(
    match.winnerId,
    winnerPrize.toString(),
    matchId
  );
}
```



Prize Distribution Flow



Tournament System

Bracket Generation

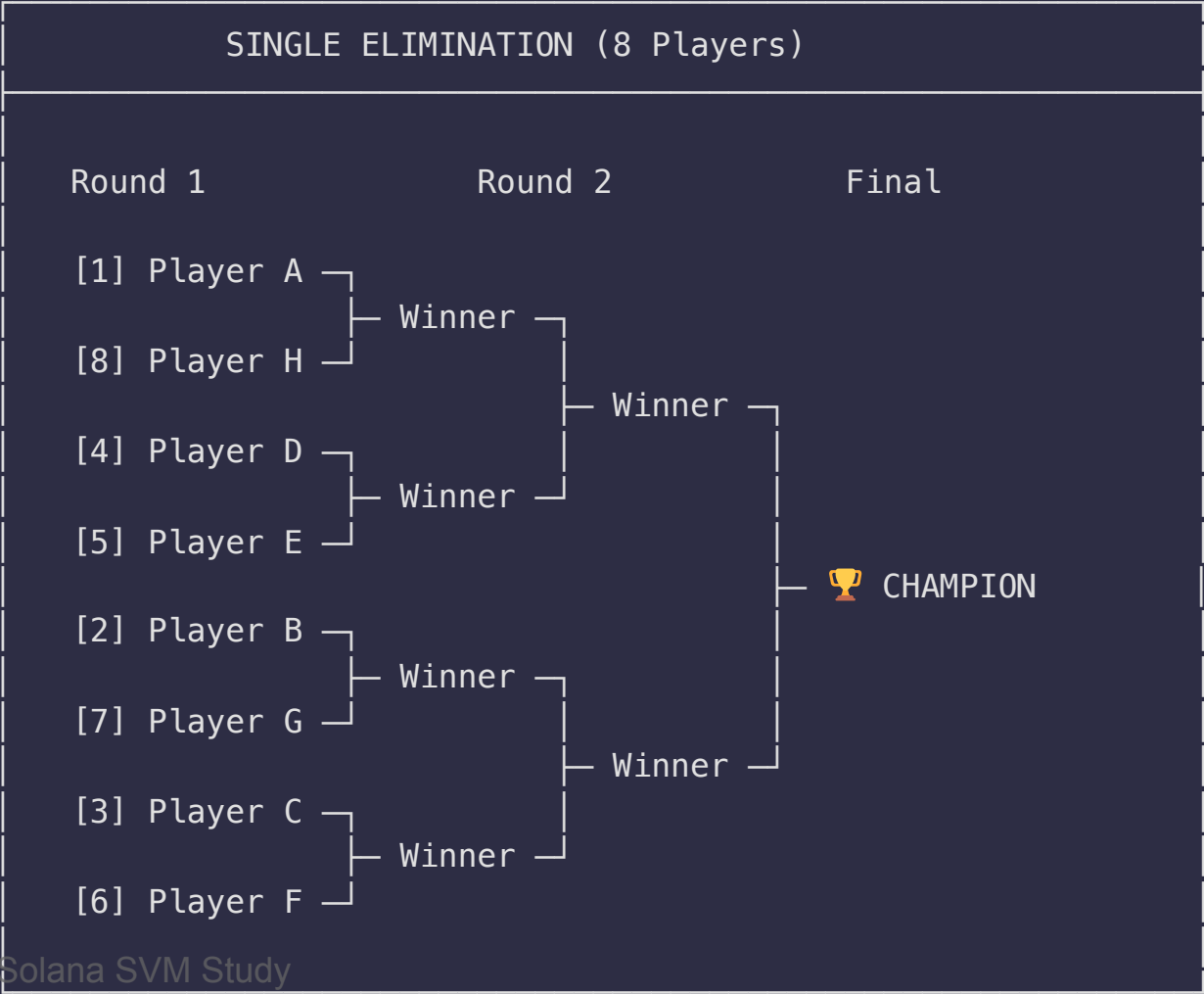
```
async generateBracket(tournamentId: string): Promise<Tournament> {
  const tournament = await this.tournamentRepository.findOne(/*...*/);
  const registrations = await this.registrationRepository.find({
    where: { tournamentId },
    order: { seed: 'ASC' },
  });

  // Single elimination: seeded matchups
  // 1 vs 8, 2 vs 7, 3 vs 6, 4 vs 5
  const bracket = this.createSingleEliminationBracket(
    registrations,
    tournament.maxParticipants
  );

  // Create match entities for each bracket slot
  const matches = await this.createBracketMatches(
    tournament,
    bracket
  );

  return this.tournamentRepository.save({
    ...tournament,
```

Tournament Bracket Structure





Tournament Prize Structure

```
const prizeStructure = {
  1: 50, // 1st place: 50% of pool
  2: 30, // 2nd place: 30% of pool
  3: 20, // 3rd/4th: 20% split
};

// Example: 16 players x 10 SOL entry = 160 SOL pool
// Platform fee (5%): 8 SOL
// Distributable: 152 SOL

// Payouts:
// 1st: 76 SOL
// 2nd: 45.6 SOL
// 3rd: 30.4 SOL
```

Placement	Percentage	Prize (SOL)
 1st	50%	76.0
 2nd	30%	45.6



Escrow Account Entity

```
@Entity('escrow_accounts')
export class EscrowAccount {
  @PrimaryGeneratedColumn('uuid')
  id: string;

  @Column({ type: 'enum', enum: EscrowType })
  type: EscrowType; // MATCH | TOURNAMENT

  @Column()
  reference: string; // match_id or tournament_id

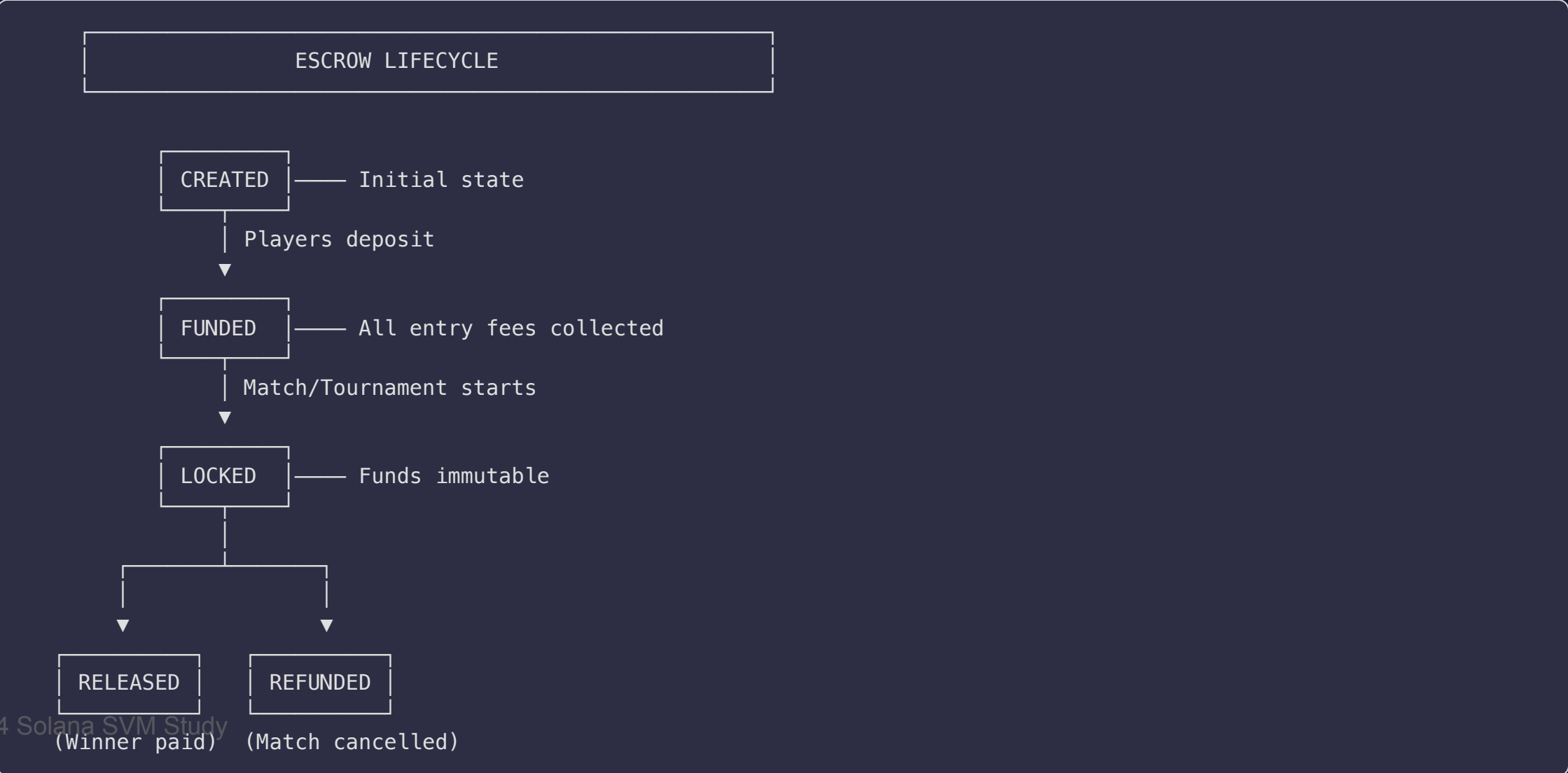
  @Column()
  escrowPda: string; // Program Derived Address

  @Column({ type: 'decimal', precision: 20, scale: 0 })
  amount: string;

  @Column({ type: 'enum', enum: EscrowStatus })
  status: EscrowStatus;
  // CREATED → FUNDED → LOCKED → RELEASED/REFUNDED
}
```



Escrow State Machine



Security Measures

Production Considerations

Wallet Security

-  MPC 2-of-3 threshold signatures
-  Daily withdrawal limits
-  Rate limiting on transactions
-  Account suspension capabilities

Escrow Security

-  On-chain PDA escrow accounts
-  Atomic state transitions
-  Immutable locked funds



Rate Limiting Implementation

```
@Entity('player_wallets')
export class PlayerWallet {
  @Column({ type: 'decimal', precision: 20, scale: 0 })
  dailyWithdrawalLimit: string;

  @Column({ type: 'decimal', precision: 20, scale: 0, default: '0' })
  dailyWithdrawalAmount: string;

  @Column({ type: 'timestamp', nullable: true })
  dailyWithdrawalResetAt: Date;

  canWithdraw(amount: bigint): boolean {
    if (this.status !== PlayerWalletStatus.ACTIVE) return false;

    const available = BigInt(this.availableBalance);
    if (amount > available) return false;

    // Check daily limit
    const dailyUsed = BigInt(this.dailyWithdrawalAmount);
    const dailyLimit = BigInt(this.dailyWithdrawalLimit);

    return (dailyUsed + amount) <= dailyLimit;
  }
}
```



Transaction Types

```
export enum WalletTransactionType {  
  // Inflows  
  DEPOSIT = 'DEPOSIT',           // External deposit  
  PRIZE_WIN = 'PRIZE_WIN',       // Match/tournament win  
  REFUND = 'REFUND',             // Cancelled match refund  
  
  // Outflows  
  WITHDRAWAL = 'WITHDRAWAL',     // External withdrawal  
  ENTRY_FEE = 'ENTRY_FEE',       // Match/tournament entry  
  
  // Internal  
  LOCK = 'LOCK',                 // Lock for escrow  
  UNLOCK = 'UNLOCK',            // Release from escrow  
}
```

Full Audit Trail

Every transaction logged with signatures and timestamps



Complete Match Flow

COMPLETE MATCH FLOW

- 1. CREATE MATCH
 - └ Host deposits entry fee → Escrow created
- 2. JOIN MATCH
 - └ Player deposits entry fee → Added to escrow
- 3. START MATCH
 - └ Escrow LOCKED → Players notified
- 4. PLAY GAME
 - └ External game server handles gameplay
- 5. SUBMIT RESULT
 - └ Winner declared → Match COMPLETED
- 6. DISTRIBUTE PRIZES
 - └ Winner wallet credited → Escrow RELEASED
- 7. SETTLEMENT
 - └ All balances updated → Match SETTLED



API Endpoints

Matchmaking

Method	Endpoint	Description
POST	/api/matches	Create match
POST	/api/matches/:id/join	Join match
POST	/api/matches/:id/start	Start match
POST	/api/matches/:id/result	Submit result
GET	/api/matches/open	List open matches

Wallets

Method	Endpoint	Description
POST	/api/wallets	Create MPC wallet



API Endpoints (cont.)

Tournaments

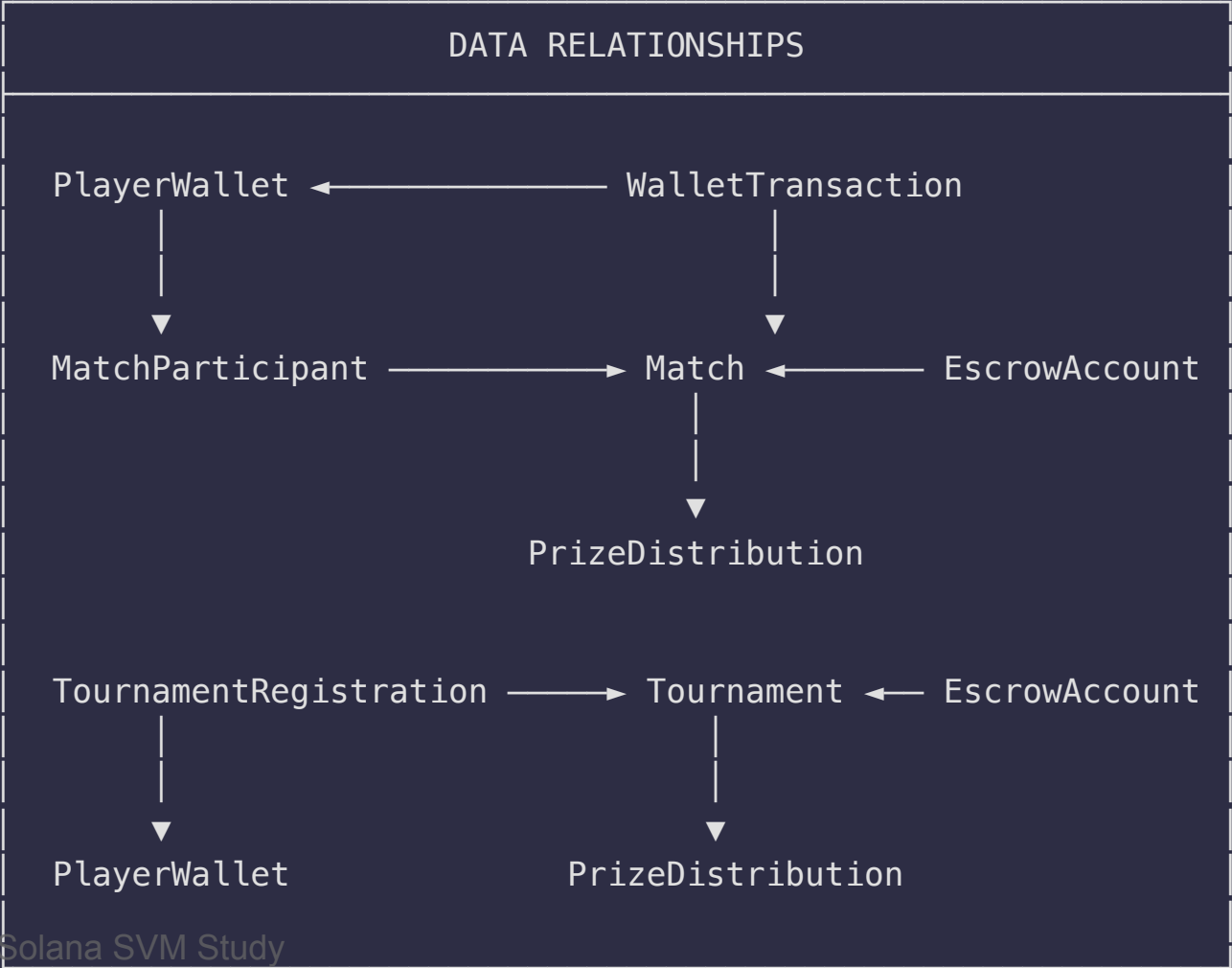
Method	Endpoint	Description
POST	/api/tournaments	Create tournament
POST	/api/tournaments/:id/register	Register player
POST	/api/tournaments/:id/bracket	Generate bracket
POST	/api/tournaments/:id/start	Start tournament
GET	/api/tournaments/:id/leaderboard	Get standings

Prizes

Method	Endpoint	Description
POST	/api/prizes/match/:id/distribute	Pay match prizes



Data Model Overview





Testing Strategy

Unit Tests

```
describe('EscrowService', () => {  
  it('should create escrow for match', async () => {  
    const result = await escrowService.createEscrow({  
      type: EscrowType.MATCH,  
      reference: 'match_123',  
      tokenMint: SOL_MINT,  
    });  
  
    expect(result.status).toBe(EscrowStatus.CREATED);  
    expect(result.escrowPda).toBeDefined();  
  });  
  
  it('should reject release for unlocked escrow', async () => {  
    await expect(  
      escrowService.releaseEscrow('escrow_unlocked')  
    ).rejects.toThrow(BadRequestException);  
  });  
});
```



Integration Testing

```
describe('Match E2E', () => {
  it('should complete full match flow', async () => {
    // 1. Create wallets for both players
    const wallet1 = await walletService.createWallet({ playerId: 'p1' });
    const wallet2 = await walletService.createWallet({ playerId: 'p2' });

    // 2. Deposit funds
    await walletService.deposit({ playerId: 'p1', amount: '2000000000' });
    await walletService.deposit({ playerId: 'p2', amount: '2000000000' });

    // 3. Create and join match
    const match = await matchService.createMatch({
      hostPlayerId: 'p1',
      entryFee: '1000000000',
    });
    await matchService.joinMatch(match.id, 'p2');

    // 4. Start and complete
    await matchService.startMatch(match.id);
    await matchService.submitResult(match.id, { winnerId: 'p1' });

    // 5. Distribute prizes
    await prizeService.distributeMatchPrizes(match.id);

    // 6. Verify balances
    const balance1 = await walletService.getBalance('p1');
    expect(balance1.availableBalance).toBe('2900000000'); // Won
  });
});
```



Monitoring & Observability

Key Metrics to Track

```
// Custom metrics for prize distribution
@Injectable()
export class PrizeDistributionService {
  private readonly prizeDistributedCounter = new Counter({
    name: 'esports_prizes_distributed_total',
    help: 'Total prizes distributed',
    labelNames: ['type', 'status'],
  });

  private readonly prizeAmountGauge = new Gauge({
    name: 'esports_prize_amount_sol',
    help: 'Prize amount in SOL',
    labelNames: ['type'],
  });

  async distributePrize(/*...*/) {
    // ... distribution logic
    this.prizeDistributedCounter.inc({ type: 'match', status: 'success' });
  }
}
```



Key Metrics Dashboard

Metric	Description	Alert Threshold
<code>escrow_balance_total</code>	Total SOL in escrow	> 1000 SOL
<code>match_completion_rate</code>	% matches completed	< 90%
<code>prize_distribution_latency</code>	Time to distribute	> 30s
<code>mpc_signing_failures</code>	Failed MPC signs	> 0
<code>withdrawal_rate_limit_hits</code>	Rate limit triggers	> 100/hr

Alerts

- 🚨 Escrow balance mismatch
- 🚨 MPC signing failure rate > 1%
- 🚨 Prize distribution backlog
- 🚨 Suspicious withdrawal patterns

Best Practices Summary

1. Security First

- Always use MPC for player wallets
- Lock escrow before match starts
- Implement withdrawal limits

2. Atomic Operations

- Use database transactions
- Handle partial failures gracefully
- Implement idempotency keys

3. Auditability

- Log all financial transactions

Future Enhancements

Roadmap Items

1. Cross-Game Wallet

- Single wallet across multiple games
- Unified balance management

2. NFT Integration

- Tournament trophy NFTs
- Achievement badges

3. DAO Governance

- Community-driven prize pools
- Voting on platform fees



Advanced Calculation Patterns

BigInt Safety for Financial Operations

```
// CRITICAL: Always use BigInt for lamport calculations
// JavaScript Number.MAX_SAFE_INTEGER = 9,007,199,254,740,991
// 1 SOL = 1,000,000,000 lamports
// Max safe: ~9,007 SOL with regular Number

// BAD: Number overflow risk
const pool = 10000000000 * 2; // Potential precision loss

// GOOD: BigInt for all financial math
const pool = BigInt('10000000000') * BigInt(2);

// Division truncates (floor), handle remainder
const prizePerWinner = distributableAmount / BigInt(winnerCount);
const remainder = distributableAmount % BigInt(winnerCount);
// Option: Add remainder to 1st place winner
```



Atomic Transaction Pattern

Database + Blockchain Consistency

```
async distributeMatchPrizes(match: Match): Promise<PrizeDistribution> {
  const queryRunner = this.dataSource.createQueryRunner();
  await queryRunner.connect();
  await queryRunner.startTransaction();

  try {
    // 1. Create prize distribution record (PROCESSING)
    const prizeDistribution = await queryRunner.manager.save(
      PrizeDistribution, { status: 'PROCESSING', ... }
    );

    // 2. Release escrow (on-chain operation)
    await this.escrowService.releaseEscrow({ escrowId, distributions });

    // 3. Credit each winner's wallet
    for (const dist of distributions) {
      await this.playerWalletService.creditPrize(dist.playerId, dist.amount);

      // 4. Update distribution status individually
      dist.status = 'COMPLETED';
      dist.processedAt = new Date();
    }

    // 5. Commit all database changes
    prizeDistribution.status = 'COMPLETED';
    await queryRunner.manager.save(prizeDistribution);
    await queryRunner.commitTransaction();

    return prizeDistribution;
  } catch (error) {
    await queryRunner.rollbackTransaction();
  }
}
```

Error Recovery Strategies

Partial Distribution Handling

```
// Prize distribution supports partial success
for (const dist of distributions) {
  try {
    await this.playerWalletService.creditPrize(
      dist.playerId, dist.amount, sourceId
    );

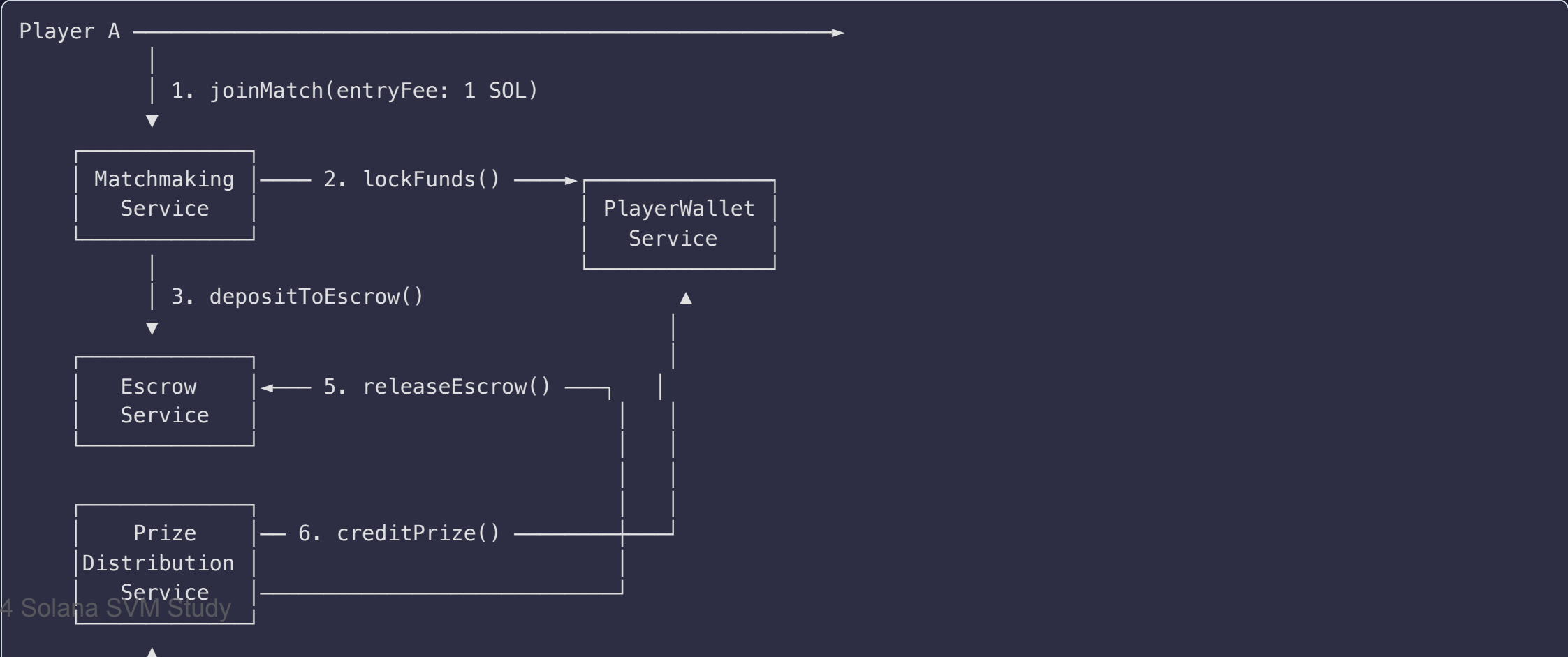
    distEntry.status = 'completed';
    distEntry.processedAt = new Date();
    totalDistributed += BigInt(dist.amount);

  } catch (error) {
    // Log but continue to next recipient
    this.logger.error(`Failed to credit ${dist.playerId}`, error);

    distEntry.status = 'failed';
    distEntry.failureReason = error.message;
    // Can be retried later via admin action
  }
}
```

Service Interaction Diagram

Complete Prize Distribution Flow



Resources

Documentation

- [Solana Web3.js Docs](#)
- [SPL Token Program](#)
- [MPC Cryptography](#)

Code References

- [/src/modules/esports/](#) - Full implementation
- [/docs/diagrams/16-esports-matchmaking.md](#) - Architecture docs
- [/src/modules/mpc/](#) - MPC wallet integration

Tools

- [Anchor Framework](#) for Solana programs

? Questions?

Key Takeaways

1. **MPC wallets** provide security without UX friction
2. **Escrow accounts** ensure trustless prize distribution
3. **Atomic state machines** prevent fund loss
4. **Platform co-signing** enables fraud prevention
5. **Full audit trails** support dispute resolution



Thank You!

Contact & Resources

-  Project Repository: github.com/psavelis/solana-svm-study
-  Documentation: </docs/diagrams/16-esports-matchmaking.md>
-  Tests: /src/modules/esports/__tests__/

 **Let's Build!**

Production-Ready Esports on Solana